

Efficiency of Panopto

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Abstract—The objective of this research paper is to investigate the effectiveness of Panopto as an e-learning tool. Due to increasing use of remote learning, Panopto has emerged as a popular platform for recording and sharing educational lectures. However, the platform’s effectiveness for students depends on its usability, which affects the accessibility and consumption of course material. Therefore, this study aims to identify where, if there is any, place where there is a significant drop in efficiency and how this could effect user experience. The research question is “How efficient is Panopto as an e-learning tool?” The paper covers the background of usability, usability testing, and the features of Panopto. The research method selected is a case study, the data collection method is observation and the data analysis method is also quantitative. The findings of the study are intended to assist designers in making better video sharing platforms by identifying patterns and areas that could use improvement.

Index Terms—panopto, usability, efficiency

I. INTRODUCTION

With the rise of remote learning, students are often required to access video lectures on their own time and from various locations. One of the platforms used for this is Panopto, which has become popular for recording and sharing educational lectures, making it a valuable tool for students. “In Panopto you can create recordings direct from your computer, record lectures in certain classrooms, and upload recordings through an integrated playback platform. Panopto is integrated in Blackboard, and recordings can be published to students simply and effortlessly.” [2]. This is how Panopto is described on the NTNU website. However, the usefulness of the platform for students relies heavily on its usability, as it directly impacts the ability to access and consume course material. Based on our usage of the platform, and the difficulties it inflicted, we deemed it an adequate subject for research regarding the sheer efficiency in the use of Panopto. By researching the efficiency of Panopto’s usability, one can identify ways to improve the platform’s accessibility, responsiveness, and overall user experience, making it easier and more efficient for students to watch lectures and engage with course content. Furthermore, improved efficiency in usability can have a positive impact on student’s academic success. By reducing the time and effort required to access and engage with video lectures, students can focus more on learning and retaining the course material, ultimately leading to better academic performance. In addition to benefiting students directly, research into the efficiency of Panopto’s usability can also help educators and instructional designers optimize their course content delivery. By identify-

ing areas where the platform can be improved, educators can ensure that their students have the best possible experience while watching lectures, leading to higher satisfaction rates and better student outcomes. This is the motivation for our research question:

RQ: How efficient is Panopto as an e-learning tool?

II. BACKGROUND

The concept of usability is based on the research paper “Defining usability” by N. O. Heaton [4], which defines usability “in a manner that can incorporate users, tasks, and environments to set explicit performance goals relating to the effectiveness, training time and subjective opinions of sample users”. For the system to be optimized for usability the user needs to be able to quickly assess the resource they want with minimum effort [5]. To test user satisfaction we can among other methods test the time they used to complete a task, if they need assistance to find information, how the user expects to be able to find their resource, the effort of which they used to search or the effort to remember steps needed to proceed. An important aspect of the paper on usability is to lay grounds for testing the product; “Essential to any approach to usability is that it must be defined so that it is testable.”

After defining what to test for, one should also define how to test. While there are many ways to test for usability, the focus of this research paper is on the effectiveness of the product. The paper by E. Wang B. Caldwell [3] compares two different usability-testing methods, heuristic evaluation, and user testing. There are different strengths and weaknesses of both methods. “User Testing can provide more insightful data from real users such as user’s performance and satisfaction.” The paper also specifies user testing as more appropriate for a product later in the development cycle, where “User Testing should be considered as a major testing method because it can capture real usability problems from the real users.” Based on this, user testing was chosen as the optimal testing method for this study.

III. METHODS

A. Research methods

Since the purpose of the study is to understand the usability of Panopto in the form of efficiency, holding a case study is the research strategy that best suits this task. The other research strategy could have been to hold an experiment on the subject. However, when holding an experiment there has

to be a hypothesis. This hypothesis should be tested through either two different groups of subjects or different variables. The research subject should be isolated from what is being tested during an experiment. Yet, the research objective is about testing a specific part of the digital tool. Even though this case study is quite small and easy to reproduce, getting the same results depends on the participants involved. This study is based on people's experiences and how much they have used the tool before. Since this is not a new tool being tested, the results of the case study can be seen as skewed. Therefore, the use of a case study is the best research strategy based approach. [6]

The data-generating method to be used is observations. This is due to the nature of the case study. The goal is to observe the amount of time participants take. This can only be done through observations as holding interviews or sending out questionnaires would not give the data that is desired. Therefore, it is possible to rule out research strategies that would use these time of data-generating methods such as surveys and design and creation. This is due to the nature of trying to understand efficiency. To measure efficiency, the subjects are given several tasks to complete. By measuring how quickly they complete these tasks, we can calculate the time-based efficiency of Panopto [7].

The data that is generated through the case study is quantitative, which is therefore the data analysis method. As the goal is the study is to test efficiency, there is a number needed to identify how efficient systems are [6]. Through this data, the objective is to see patterns and areas where there are problems. The main threat to testing efficiency is how well the subjects know the system beforehand, as this system has been in use for quite some time at NTNU some subjects could have already frequently used it. However, even with that being the case participants can still struggle with tasks that may be outside of their use area for the website.

B. Participants

As our area of research is based on an educational e-learning tool, the participants are to be students at a high education level. Since, as of now, the only access is to the viewer version of Panopto and not the publisher version, there is a restriction on the types of participants in our case study. The participants in the case study are our group as facilitators and observers and a group of students as subjects. These students can come from any background as the goal is to test efficiency regardless of technological abilities. The differences in types of subjects are necessary as different technological backgrounds may affect the understanding of technological websites. As this case study has a narrow scope, the general aspects can still be utilized by others trying to make a video platform for lectures. This research can either be used by Panopto's user interface team, or other video providers wanting to make an efficient platform.

Our role as facilitators is to be almost invisible. We are there to observe the subjects while they perform the tasks given to them. The goal of facilitators is to not help in any way. Especially since the case study is based on efficiency,

whatever mistakes they make or however long it takes them to complete the test is very relevant to the total time tasks take. This means that if we were to assist in tasks, the data we are gathering would be tampered with, resulting in data that gives no value. Therefore, our role is to observe and not assist, but to make sure that the tasks given are not leaked. Leaked tasks could also result in the subjects practicing the tasks and performing better than they would have. As all of us are technology students it is important that the tasks given during the experiment are general enough to test the different abilities of Panopto. There would be little point to try and get bad results by making difficult tasks and that would not contribute to making efficient video platforms. Therefore, we as researchers have to be unbiased and neutral when conducting the case study so as not to influence the test subjects in any way.

Since we are performing a case study, it would be helpful to record the computer screen to accurately time the completion of tasks. Therefore, there is little need to think about anonymity and confidentiality as the only thing we are recording is the computer screen, as long as the subjects are comfortable sharing their technological abilities anonymously. This is not data that can be traced back to people as it would only show the difference in technological abilities. To make sure that the screen recordings do not somehow capture personal data from subjects, we will be providing computers to the subjects in these tests. [6]

C. Research Paradigm

Positivism is the research paradigm most suited for this research study. This is due to the basic assumptions that the world is ordered and regular and it can be investigated objectively [6]. As we the researchers are a neutral part of the case study, our objective is just to observe the subjects and gain quantitative data from their performance, we fall under the characteristics of positivism. This is not the case for all case studies, however with the nature of this subject, as it collects quantitative data, it can be placed under the positivism umbrella.

D. How the Case Study is Conducted

The case study will be held in a room with a computer setup to Panoptos front page as that is the starting point of every task. The participants in the case study are at least two researchers, one being the facilitator and the other being an observer, and a participant being the test subject. Before the case study starts the facilitator will inform the subject of their right to withdraw at any moment and that all information will be anonymous. The only recordings that will be taken are those of the screen that the experiment is being held on. The facilitator will have a stopwatch where they record the time it takes for each subject to complete the tasks and write down this time. Before the case study starts the facilitator asks the participants "How would you rate your technology skills on a scale from 1 to 5?", this could help give an understanding of the different times that participants have. As having good technological skills can help

when performing tasks on a computer. The facilitator will ask the subject to complete the following five tasks.

- 1) Find the first recording from the lecture in the subject TMA4100 from fall 2022. (TMA4100 Matematikk 1 - Informasjon om Matematikk 1 (del 1))
- 2) Find the second recording from the lecture in the subject TMA4100 from fall 2022. (oversiktsforelesning 1)
- 3) You have been sick and missed the lectures from week 44, find them in the subject TMA4100 from fall 2022. (This is from the 31 of October to 6 of November) (oversiktsforelesning 10)
- 4) Find and subscribe to NTNU Trening. From subscriptions find the longest yoga session from NTNU Trening (yoga 31. March)
- 5) Unsubscribe from NTNU trening

By having subjects go through these questions, we test the different abilities that Panopto has to offer. As the goal is to test efficiency, there have to be generic tests of the different tools to test how efficient they are. The goal of the case study is to gather quantitative data, where the focus is on the amount of time tasks took. After all the data is collected from the participants, it will be gathered and analyzed to see what can be concluded from the results.

IV. RESULTS

Before each case study, the participating individuals were asked to rank their technological experience on a scale of one to five, with one being completely inexperienced, and five being expert-level. Upon reviewing the results after each individual had been tested, it became clear that all participants thought of themselves as well-versed in the world of technology. They either rated themselves a four or five on the experience scale.

As illustrated in Table 1, the participants and their respective test results were sorted in rows based on their alleged level of competence. To summarize the case study process, each participant was presented with five tasks to perform on Panopto, which is a service used mainly for streaming lectures online and/or storing it for revision later on. The study found that students with a supposed top level experience performed only marginally better than those who rated themselves slightly lower.

Panopto is intended for all kinds of people regardless of their background. Considering this, it is evident that there is a lack of efficiency in usability of this site when even some of the most competent test subjects struggled to the point where it took them five minutes to finish a seemingly simple task. The purpose of this experiment was to see if Panoptos features actually are effective to the extent where students can find their lectures and other course material online.

We found that students with lower competence levels had a steeper learning curve when advancing from the first task to the next one. However, this was not the case among the participants who rated themselves higher, as they did not experience the same progression and spent more time completing the next task than their less competent counterparts.

TABLE I
SHOWS THE TIME USERS USED ON THE DIFFERENT TASKS GIVEN TO THEM. THE TASKS CAN BE FOUND IN SECTION III-D.

Subject	Task 1	Task 2	Task 3	Task 4	Task 5
Student 1 (5/5)	00:23.30	00:24.00	00:10.00	1:42.50	00:15.00
Student 2 (5/5)	00:09.00	01:21:10	00:50.00	02:35.00	00:13.00
Student 3 (4/5)	01:41.50	00:15.00	00:20.00	05:24.00	00:36.50
Student 4 (4/5)	00:19.20	01:26.20	00:26.00	01:06.20	00:13.50
Student 5 (4/5)	02:00.00	00:45.00	00:20.00	02:06.00	00:01.00
Student 6 (4/5)	02:01.00	00:13.46	00:22.31	02:03.51	00:09.06
Student 7 (4/5)	00:59.35	00:14.03	00:15.21	01:58.67	00:10.71
5/5 experience	00:16.15	00:52.55	00:30.00	2:08.75	00:14.00
4/5 experience	01:14.21	00:34.73	00:20.70	2:31.67	00:14.15
Students	00:59.19	00:39.80	00:23.36	02:02.26	00:14.03

In case studies of this nature, it is important take note of the variables at play. Accordingly, when a participant claims their technological skills to be at a certain level, it is difficult to determine whether this is factual or simply a personal opinion. However, there is no denying that the participants with a higher rating performed better overall when compared to the ones who rated their own competence slightly lower.

The bottom three rows of Table I show the average task completion time across different groups, and reveals that the fourth task of the experiment was significantly more time consuming than the others. To put it in perspective, the task with the fastest average completion time across all groups was the fifth one, at 00:14:03. That's almost eight times faster than the former average.

The penultimate row and the one third to last reveal that most of the tasks had a similar completion time on average. Despite the difference in experience level, both parties struggled roughly the same amount. Thus it is rather clear that plays little role when comparing the performance of individuals with differing levels of experience when it comes to technology.

V. DISCUSSION

During this case study the results that were obtained reflect how we originally perceived Panopto as a video-sharing application. The data as seen in table I shows how the test subjects struggled when trying to find certain videos and folders. As the application is used often by students to find videos and resources for an educational setting, they should not be hindered its inefficiency. This case study specifically tested each user from the home page of Panopto. However, this is most likely not how the users would usually have used the website. Most professors in subjects would link to a specific video for each lecture that they perform, there reducing the overall contact with the different features that NTNU Panopto has to offer. This gives the users more control over what video they are accessing but does not contribute to the understanding and improvement of the video-sharing platform. If users constantly have to go to a different website to access videos within the Panopto site, it shows that the page is not built with usability in mind and not a tool that therefore should be used in the higher education space.

When doing the efficiency test, a good number of the users rated their technological abilities to be either a four or a five. This type of score would indicate that they view themselves as highly technological people, with a good understanding of how websites should function. Despite this, most of the users struggled with the tasks presented to them. Therefore, showing that usability and specifically efficiency could not have been a focus during the production of the page. Efficiency is important for webpages as unnecessary steps and extra time that tasks take make users less inclined to continue using the webpage. Especially if they start to get frustrated when trying to perform tasks. This could result in few users and less business for Panopto. Which of course is not in Panopto's best interest.

NTNU's description of Panopto mentions its ability to produce and distribute videos to students. It does not however comment on its user interface and how easily it is for users to map through its website. As per definition of usability, Panopto as a e-learning tool should allow users to quickly acquire specific resource at a minimum effort for it to be considered to be optimized in terms of usability.

As user testing was applied for efficiency quantification the results in table I provides a concrete evaluation of its usability. Some tasks were performed quickly using the integrated search bar, however as tasks were made to have multiple steps involving time specific filtering, mapping and subscription to Panopto accounts even students of ample self proclaimed computer skills struggled. Task 4 introduced up to 5 minutes of work for students to subscribe to a specific user and locate its longest video.

When performing the tasks some issues arose. As the purpose of the study was to test efficiency on Panopto, we wanted to let the test subjects try to perform as well as they could until they felt done. This is how it would be in real life, as a user would not always know the name of a specific video but rather look at the date and work from there as to what the content is. This, therefore, caused some problems during testing as the test subjects found themselves to be done, but they had not found the correct video that was wanted by the team. Having this issue arise with multiple different test subjects, it became clear that the problem was with the wording of the task as well. Here however is difficult to make tasks that are specific enough to test the efficiency, making them too specific as to almost tell the users what to search for. This was a difficult task as finding a middle ground that would best fit this research was difficult and could have been improved upon. However, as testing had already started we decided to continue with these sets of tasks to see if this was an overall problem.

Another issue arose, as there was more than one lecture in week 44. The task just said to find a lecture from that week. This caused different participants to find different lectures from that week. This was due to the task not specifying which lecture the users should find. In real life, if a user was sick for a whole week they would need to find the first lecture of that week and work their way from there, but as this was not

specified individuals found themselves feeling like they had completed the tasks before they had found the lecture that we had anticipated that they should find. This along with other complications make it so that the data obtained on average shows less time than the tasks would have taken if they knew specifically what lecture they were looking for. Overall the time tasks took far exceeds the amount of time you would expect these types of tasks to take, therefore showing a lack of efficiency in Panopto as an e-learning tool.

VI. CONCLUSIONS

Panopto works in general for sharing educational lectures, but is not optimized in regards to time usage on the website, and has issues being used by itself instead of through other sources. With a focus on its usefulness, we delved deeper into understanding the usability of the platform and how that can be measured. We chose a case study as that would be more beneficial for the results, which were gathered by observing the participants finishing our set tasks. Observing the results we can see that technically experienced participants struggle to complete what we imagined as normal situations a user might encounter when using the website. By optimizing this interaction and focusing on improving the flaws we focused on, in return we believe the usability of the website will increase, leading to a higher satisfaction rate for all students and educators alike.

The next steps in terms of research would be to use this data and try to improve upon the existing system. This research is very specific for Panopto as a website, so it would be most useful for Panopto to continue the work with improving the usability. However, this can be looked at in a general way as well. Since the tasks that were chosen for testing were generic in how such a system should work, the goal for other video sharing platform should be to have systems that are more efficient when trying to find videos and such. Overall, this study could be used by anyone, but there would need to be more usability testing to make sure that those new systems give users a good experience.

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